

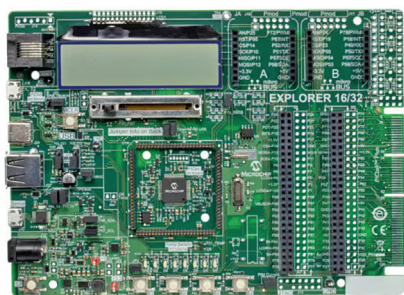
Applications. This reference design helps you validate, prototype, and deploy industrial sensor applications faster.

Link. <https://www.electronics-foru.com/electronics-projects/industrial-multi-sensor-reference-design>

OEM Brand. STMicroelectronics

Advanced sensor interfacing

Modern automotive and industrial systems use more sensors every day, and when we design these systems, we face the challenge of handling sensor signals, adding safety mecha-



nisms, and meeting industry standards. This design from Microchip helps us tackle that challenge by giving us a CPU with a built-in DSP engine so we can process signals in real time and run signal conditioning algorithms while using integrated analog, safety, and security features. The reference design shows how we can implement sensor control following ISO 26262 functional safety steps, guiding us through the full design flow from hazard analysis and safety goal definition to adding hardware and software safety mechanisms. By including a subset of safety features used in real applications, it gives us a starting point to build compliant systems without starting from scratch.

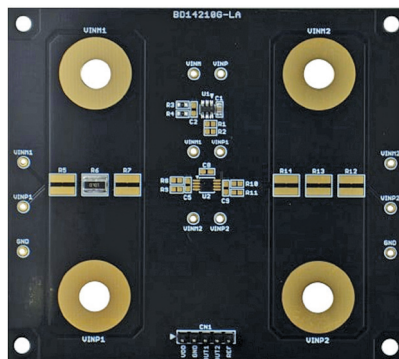
Applications. The reference design can be used for automotive sensor systems, industrial automation, safety-critical systems, and real-time monitoring.

Link. <https://www.electronicsforu.com/electronics-projects/reference-design-for-advanced-sensor-interfacing>

OEM Brand. Microchip

Current sensing solution

When we design circuits, measuring current can be tricky. We know offset errors, bias currents, or mismatched gain resistors can throw off our results, and we still have to balance accuracy, board space, and reliability. That's where this reference design from ROHM helps us. It



combines a current sense amplifier with a shunt resistor in a ready-to-use design, letting us handle high-side or low-side sensing, motor control, or overcurrent detection without worrying about pairing or tuning components. We can connect it directly to an AD converter or integrate it into a larger system. It gives us 1% current sense accuracy, handles measurements up to 30A, and its layout saves board space. With this design, we spend less time on trial and error and more time focusing on the system we want to build.

Applications. The reference design is suitable for industrial or space-constrained applications.

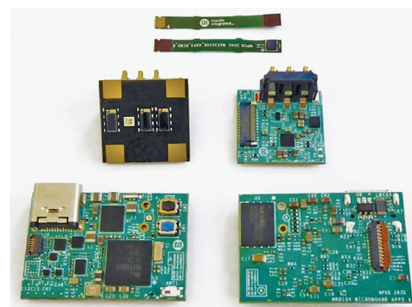
Link. <https://www.electronicsforu.com/electronics-projects/reference-design-of-a-current-sensing-solution>

OEM Brand. ROHM

Health sensor platform

We often spend months developing accurate sensors for wearable health devices, only to face delays in testing and clinical trials. Collecting reliable data for SpO₂, ECG, heart rate, and other vital signs while keeping power low adds to the challenge. With the reference design from Analog Devices, we can skip the long setup

and get straight to testing. It's a ready-to-use platform that tracks SpO₂, ECG, pulse rate, heart rate, body temperature, and motion, with built-in algorithms for HR, HRV,



respiration, SpO₂, sleep quality, and stress at clinical accuracy. We can use it directly in prototypes or trials and adapt it easily for wristbands, chest patches, or smart rings, saving us months of design time.

Applications. The reference design helps us measure pulse transit time, making it useful for cardiac health applications.

Link. <https://www.electronics-foru.com/electronics-projects/reference-design-of-health-sensor-platform>

OEM Brand. Analog Devices

To conclude

Each reference design eliminates the lengthy trial-and-error phase, enabling immediate progress to building and testing. Whether developing ultrasonic sensors for robots, smart industrial nodes for factory networks, safety-compliant systems, precision current monitors, or wearable health devices, these ready-to-use blueprints centre attention on performance and innovation rather than starting from scratch. Leveraging proven designs from trusted brands allows faster development, lower risk, and more efficient delivery of reliable sensor solutions to real-world applications. **EFY**

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